CyberForge

A Real-Time Agentic AI System for Web Threat Detection and User Protection

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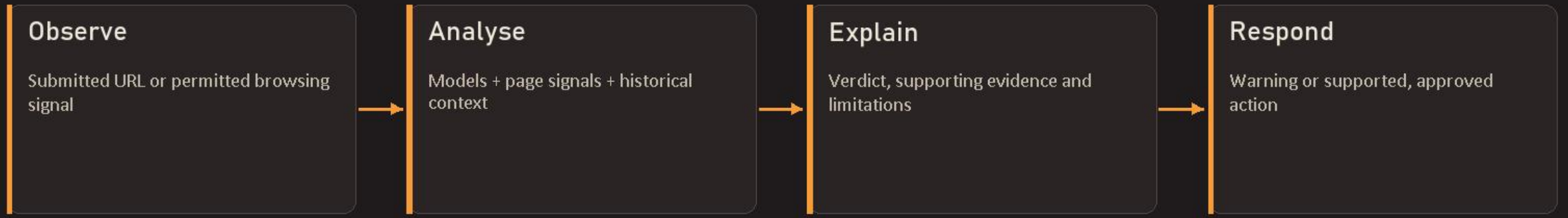
Supervisor: Prof. ONABID MATHIAS AKONG



RESEARCH + CURRENT SYSTEM
Source review · 18 October 2026

Brief introduction

CyberForge: real-time web-threat detection through cooperating specialist agents.



Research idea: combine specialised analysers into one inspectable security workflow.

Problem statement

Existing approaches struggle with false alarms, unclear verdicts and fragmented context.

01 / TRUST

Too many false alarms

Biased scores interrupt safe browsing and encourage alert fatigue.

02 / EVIDENCE

A score without a story

A single verdict does not explain indicators, behaviour or attack techniques.

03 / CONTINUITY

Disconnected context

Past observations and threat intelligence are difficult to reuse consistently.

Thesis §1.2; original calibration described in §3.7.2

Research questions

General question: real-time, explainable web-threat analysis across interfaces.

01	Decomposition	How should URL analysis be divided, and what agent set is sufficient?
02	Integration	How can URL, content, IOC, ATT&CK and reputation signals agree?
03	Decision	How can aggregation reduce false alarms without hiding strong threats?
04	Context	How can past browsing observations support future assessment?

Thesis §1.3; validity qualifications in revised §§3.9 and 4.9

Research objectives

Build and evaluate an explainable, multi-platform web-security system.

Coordinate agents

Decompose URL analysis into ordered, parallel tasks.

Unify evidence

Combine model, structural, content and behavioural signals.

Control false alerts

Balance aggregated evidence with strong threat indicators.

Add threat context

Link findings with attack techniques and external intelligence.

Brief literature review

Prior work motivates hybrid detection, structured explanations and agent coordination.

Khonji et al. · 2013

Hybrid signals

URL, page content and reputation offer complementary evidence.

Le · 2018 / Devlin · 2019

Learned representations

URLNet and BERT reduce reliance on manually selected patterns.

Strom et al. · 2018

Structured explanation

ATT&CK gives findings a shared vocabulary of attack techniques.

Wooldridge · 2009
Yao et al. · 2023

Agent coordination

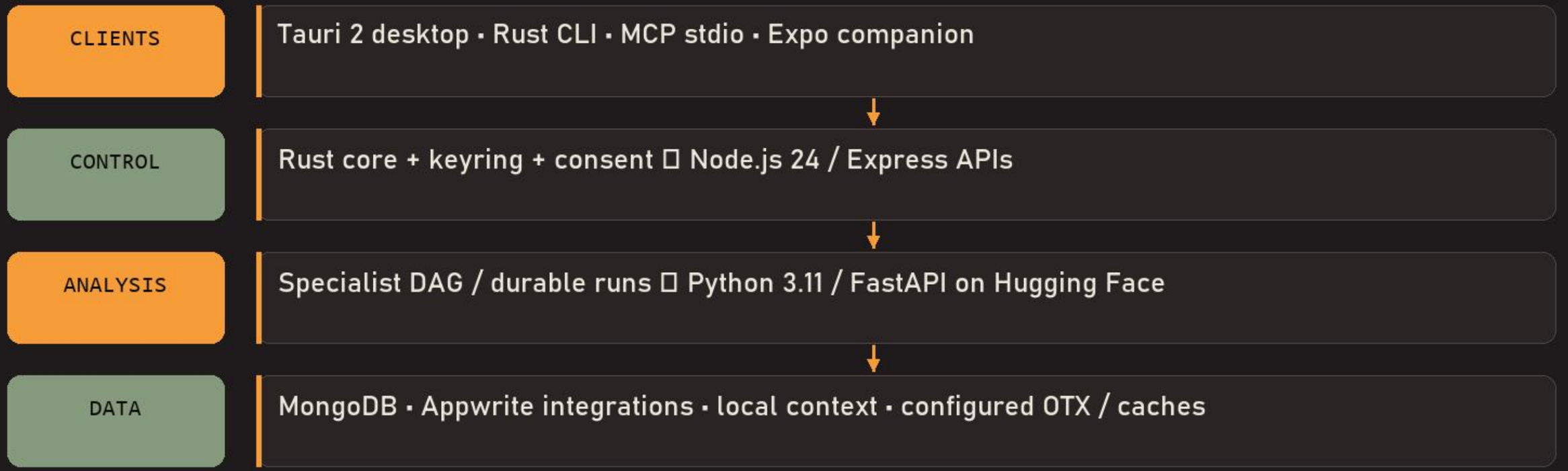
Specialisation and tool-based reasoning motivate modular workflows.

Research gap: reproducible integration, meaningful failures and usable evidence.

Thesis Chapter 2; full bibliographic details are in the speaker notes

Methods and materials used

1 of 4 · Software materials and system architecture

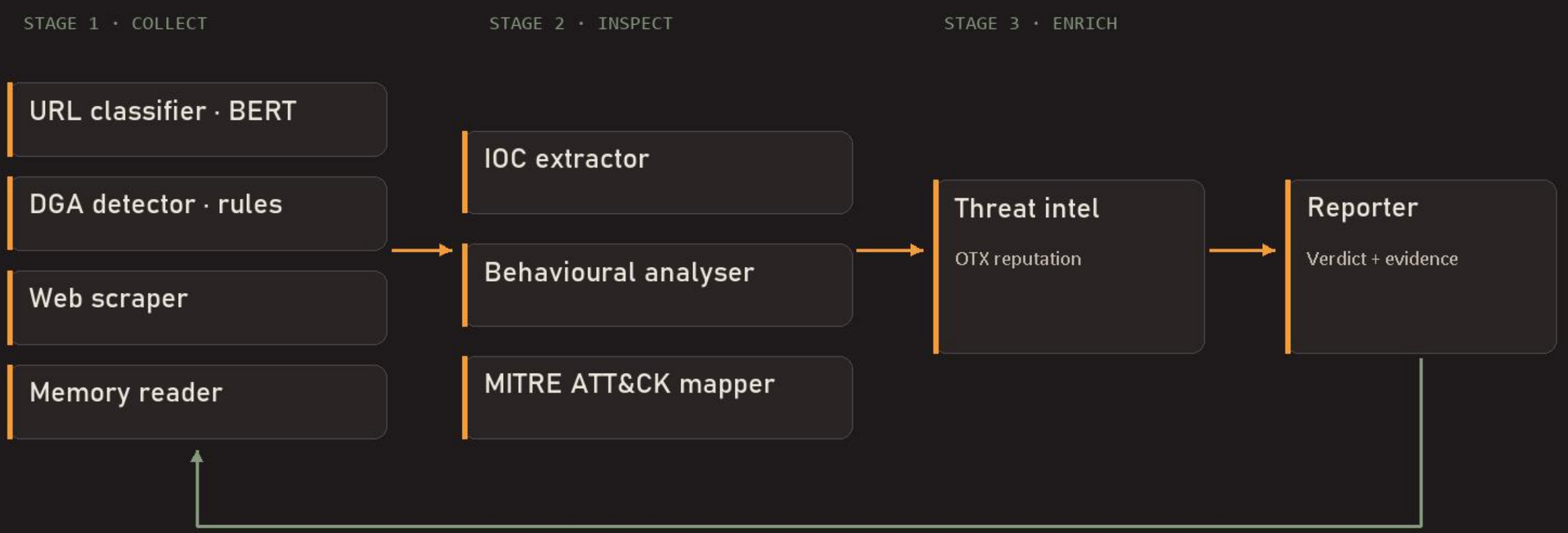


Identity, consent and evidence cross every layer; model availability is reported separately.

Revised §3.2 and Appendix C; Cargo.toml, backend/package.json, hf_space_deploy/

Methods and materials used

2 of 4 · Eight-agent analysis pipeline and coordination method



Persisted context supports later visits — it does not retrain the models.

Thesis Table 3.1; backend/src/services/agentOrchestrator.js

Methods and materials used

3 of 4 · Learning techniques and model-training phases

SUPERVISED

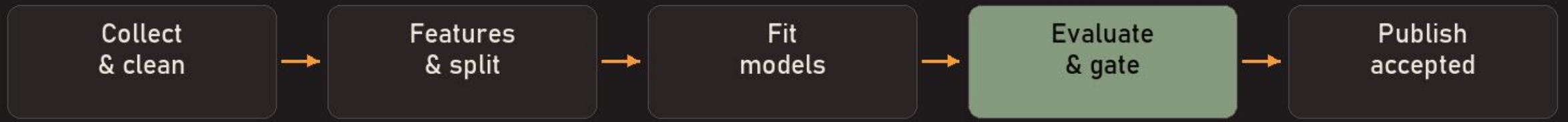
3 gradient-boosted
URL classifiers

UNSUPERVISED

IsolationForest
trained on benign URLs

PRETRAINED + RULES

BERT / Qwen inference
DGA / IOC heuristics



Domain-held-out evaluation • Shared training/serving features • No automatic online weight updates

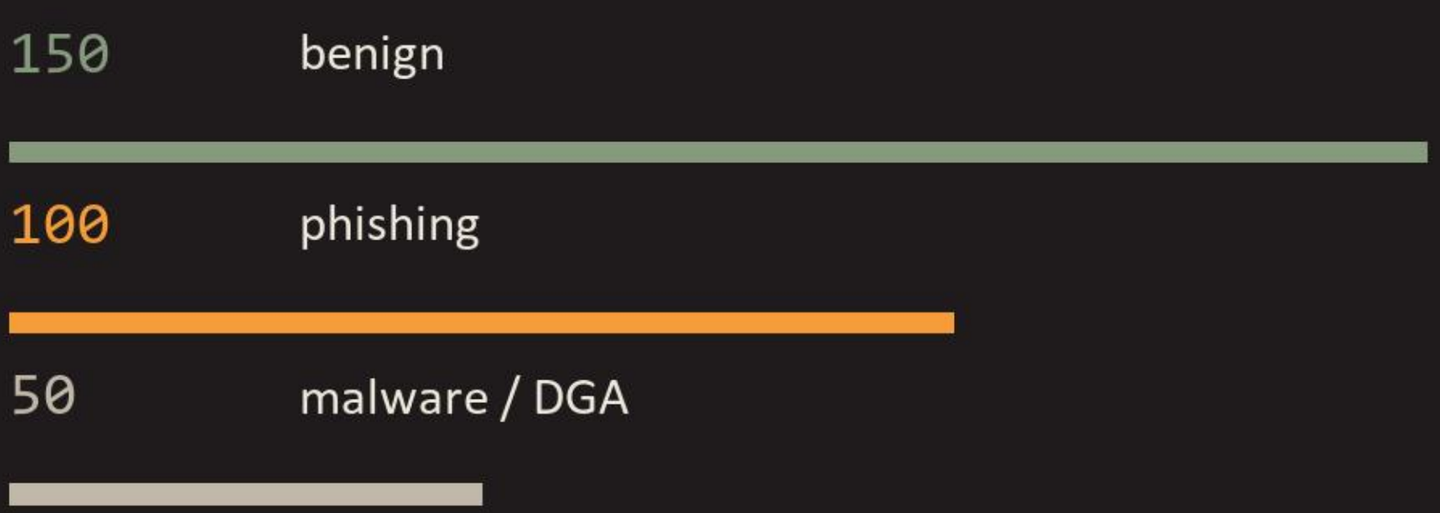
Revised Appendix C.3–C.4; hf_space_deploy/trainer.py, app.py, url_features.py

Methods and materials used

4 of 4 · Evaluation data, calibration and testing procedure

300 URLs

Historical classification evaluation set



Development calibration
7 URL examples + 30 domain labels

Current source verification
Contracts, access, failure and UI behaviour

Thesis §§3.7–3.9, Table 4.3; revised Appendix C.5

Results

1 of 4 · Detection performance: 284 of 300 URLs classified correctly

	PREDICTED THREAT	PREDICTED BENIGN
ACTUAL HARMFUL	137 True positives	13 False negatives
ACTUAL BENIGN	3 False positives	147 True negatives

94.7% Accuracy

97.9% Precision

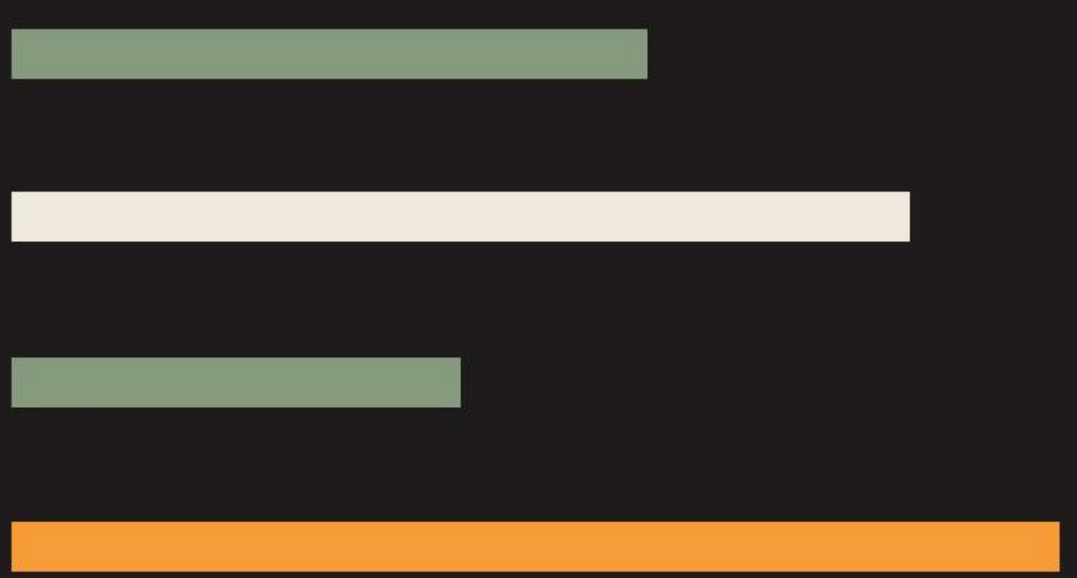
91.3% Recall

2.0% False-positive rate

Thesis Tables 4.3–4.4; confusion matrix derived directly from Table 4.3

Results

2 of 4 · Processing speed: 4.05-second mean end-to-end analysis



4.05 s

Mean total

<100 ms
cached repeat

Strict warm target: 4.00 s → measured mean: 0.05 s above target.

Thesis Tables 4.1–4.2 and §4.6; original warm target in §3.9

Results

3 of 4 · Explanation quality and the user-facing alert

Factual accuracy 9.5/10

Readability 9.0/10

Useful advice 8.8/10

Human-rated, descriptive results.
Not a blinded user study or a current-model score.



Original thesis interface illustration

Thesis Table 4.6 and Fig. 4.3 (original alert screenshot)

Results

4 of 4 · Findings in relation to the research objectives

RESEARCH AIM	REPORTED FINDING	INTERPRETATION
Coordinate specialist analysis	Eight specialists + reporter	Implemented architecture
Classify URLs accurately	94.7% accuracy	284 / 300 correct
Reduce false alarms	2.0% false-positive rate	3 / 150 benign URLs flagged
Provide timely feedback	4.05 s mean response	0.05 s above warm target
Explain decisions clearly	9.0 / 10 readability	Descriptive manual rating

Historical evaluation findings; no new benchmark is claimed for the revised implementation.

Thesis Table 4.8, checked against Tables 4.3–4.6 and §3.9

Conclusion

CyberForge integrates threat analysis, supporting evidence and understandable warnings.

94.7% accuracy / 2.0% false alarms

Reported on the original 300-URL evaluation.

Contribution	Specialised analysis + structured evidence + usable explanations
Next validation	Independent data, per-detector errors and controlled comparisons
Next delivery	Verified releases, training isolation and bounded AI integrations

